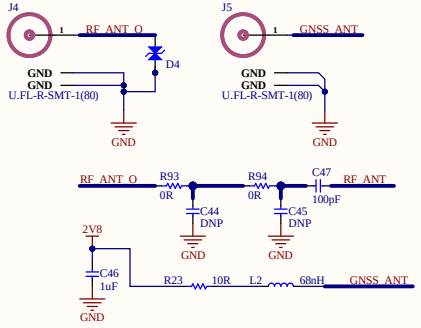


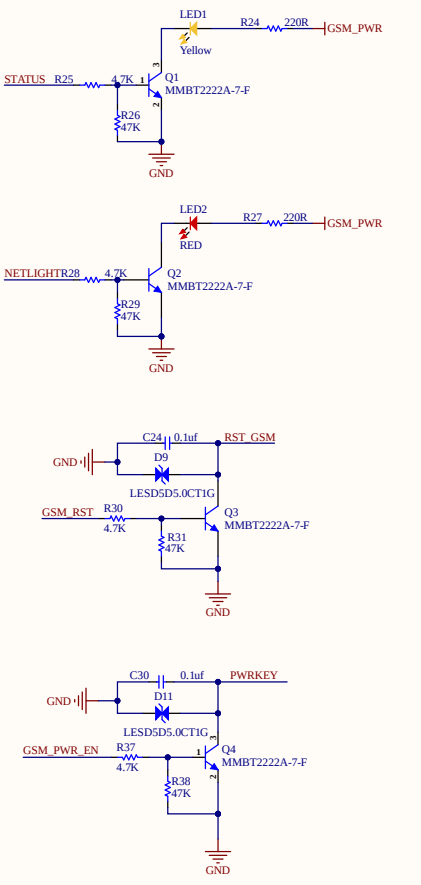
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Date: 7/17/2025	Page	Num: 1
Drawn By: Huzaifa Ali	Page	Total: 4

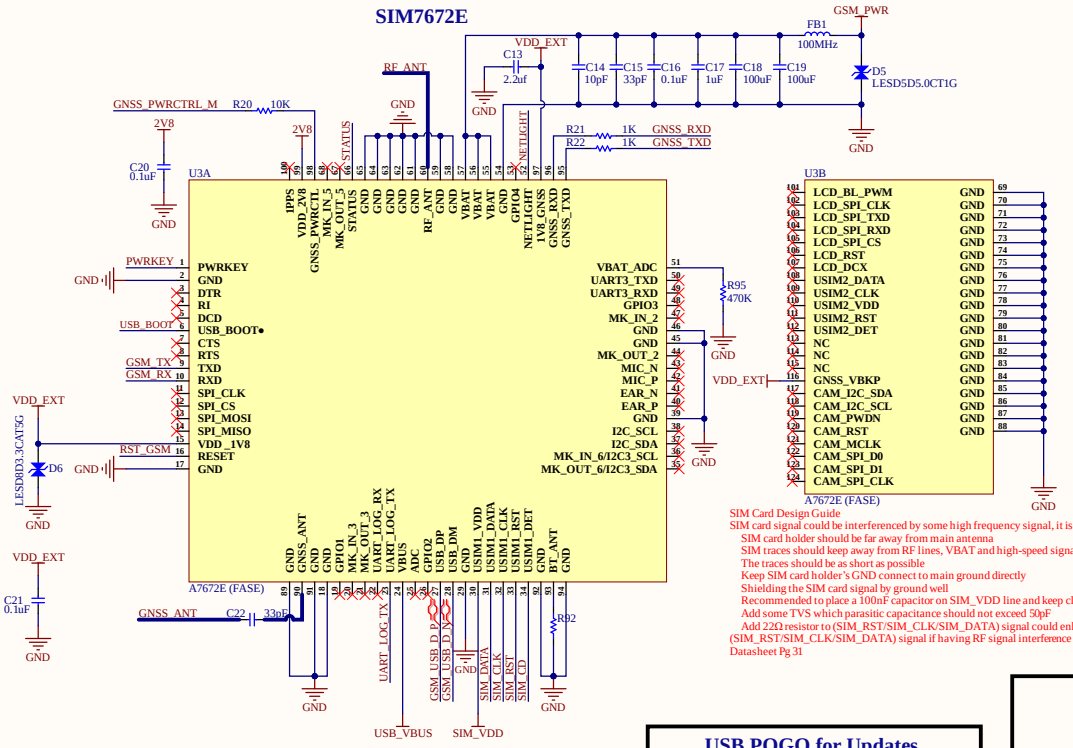
ANTENNA



GSM_COMMANDS



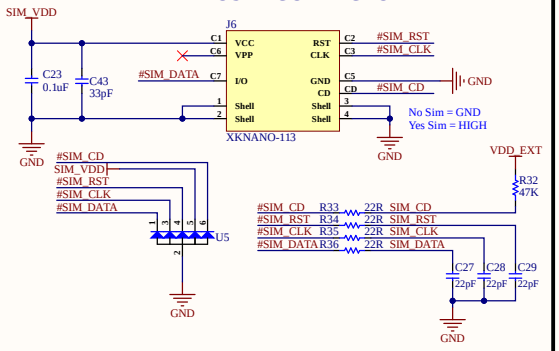
SIM7672E



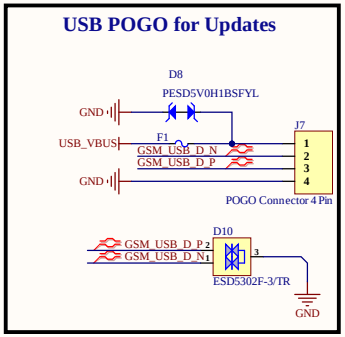
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

Datasheet Pg 31

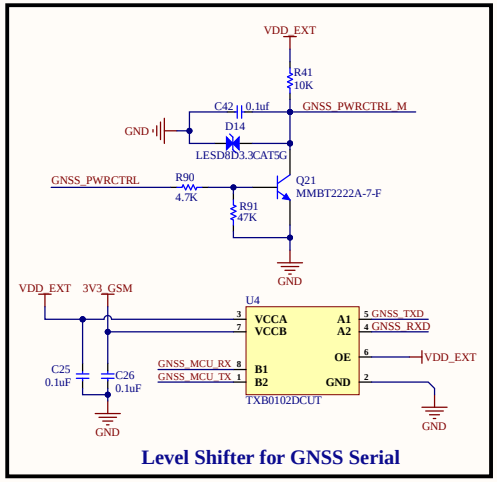
NANOSIM CONNECTOR



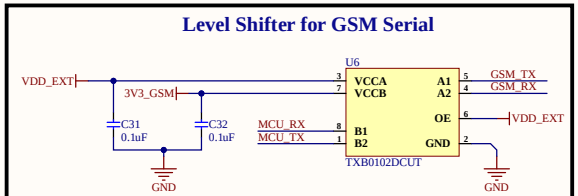
USB POGO for Updates

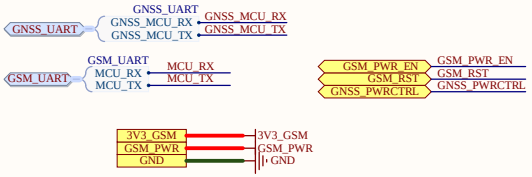


Level Shifter for GNSS Serial



Level Shifter for GSM Serial

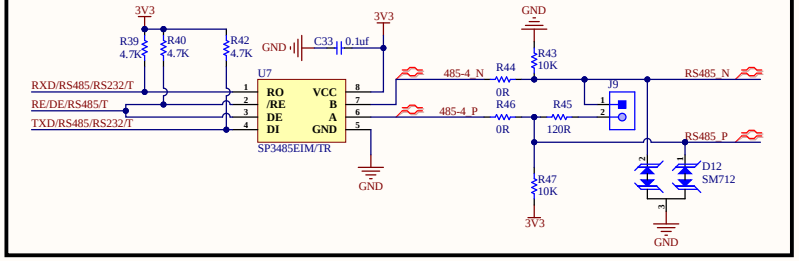




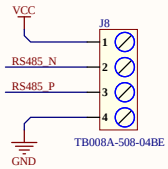
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RS485 to UART

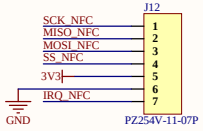


ULTRA-SONIC SENSOR



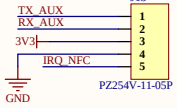
Part Selected:
https://www.massa.com/industrial/ultrasonic-sensors/#1501873783525-5657d00f-07c9

NFC SPI INPUT



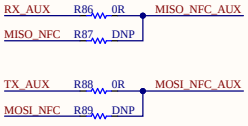
PN532 NFC Module 13.56Mhz

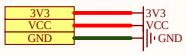
NFC UART INPUT

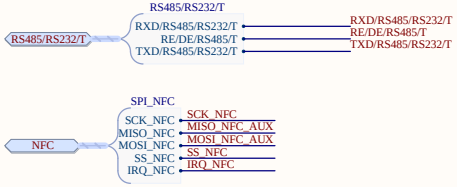


PN532 NFC Module 13.56Mhz

NFC MODE SELECT

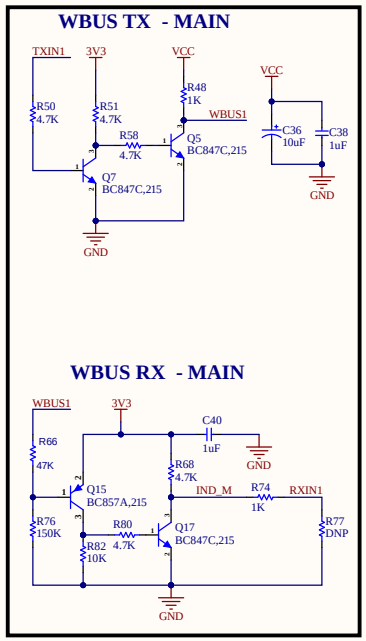


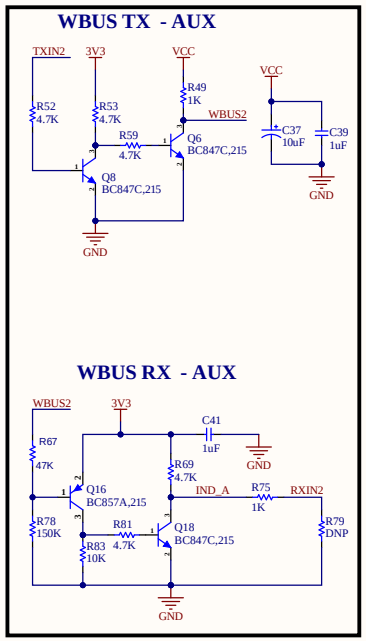


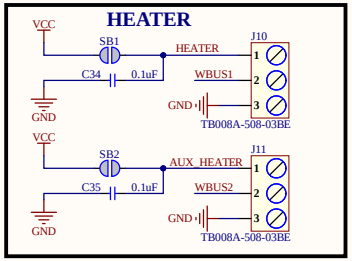


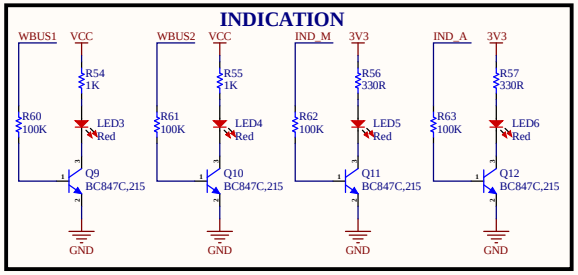
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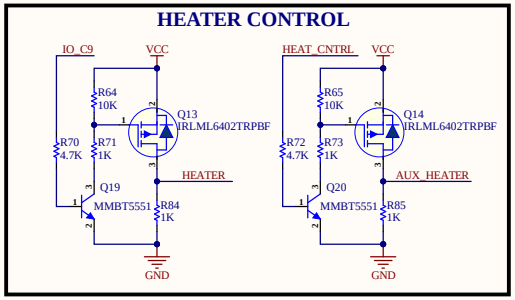
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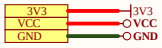










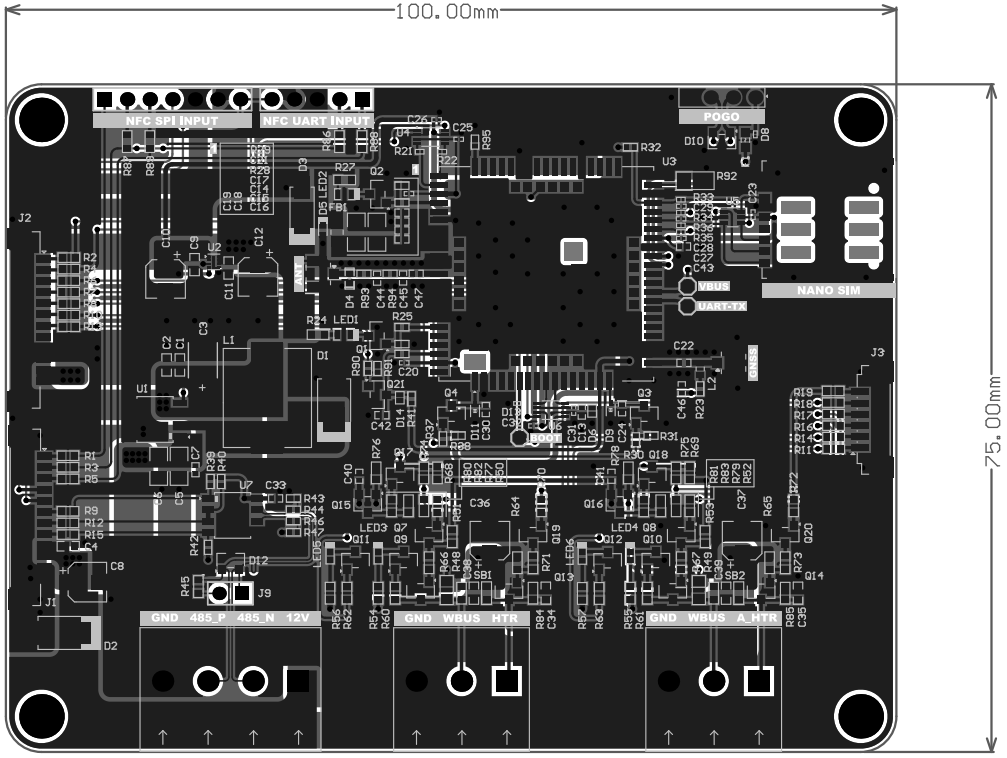






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Title: Heater_Sheet.SchDoc	
File: Smart_Connect_Board.PrjPcb	
Date: 7/17/2025	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 4



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.025mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 1	FR-4	1.500mm	4.6	
2	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.025mm	3.5	
	Bottom Overlay				
Total board thickness:			1.621mm		

Comment	Description	Designator	Footprint	LibRef	Quantity
0.1uF	CAP CER 0.1UF 50V 10% 0603	C1, C2, C4, C9, C11, C34, C35	Cap-0603	CC0603ZRNPBBN104	7
100uF	CAP TANT 100UF 20% 10V 1411	C3	Case-B	A-3810TM010KNJ	1
47uF	CAP CER 47UF 25V 20% 1206	C5, C6	Cap-1206	GRM32ER61E476KE15L	2
0.1uF	CAP CER 0.1UF 25V X7R 0402, CAP CER 0.1UF 50V X5R 0402, CAP CER 0.1UF 100V X5R 0402	C7, C16, C24, C30, C33, C42	Cap-0402	GRM155R71E104KA01D, C1005X5R1A104K050BE, GRM155R61A104KA01D	6
10uF	CAP AL/M 10UF 35V	C8, C10, C12, C36, C37	NXT1E100MD40S_	NVT1E100MD40S	5
1.2uF	CAP CER 1.2UF 25V 0603	C13	Cap-0402	GRM155R61E225KA11	1
33pF	CAP CER 33PF 50V C0G/NP0 0402	C14	Cap-0402	GRM1555C1H330JA01	1
33pF	CAP CER 33PF 25V C0G/NP0 0402	C15, C22, C43	Cap-0402	GRM1555C1E330JA01	3
1uF	CAP CER 1UF 16V X5R 0402, CAP CER 1UF 10V X5R 0402	C17, C40, C41, C46	Cap-0402	CC0402KRX5R7BB105, CC0402KRX5R6BB105	4
100uF	CAP CER, X5R, 10 V, 0.1UF [1210 Metric]	C18, C19	Cap-1206	C3225X5R1A107M/TR4K	2
0.1uF	CAP CER 0.1UF 25V 0402	C20, C21, C23, C25, C26, C31, C32	CAP-0201	GRM033C81E104KE14D	7
22pF	CAP CER 22PF 50V C0G/NP0 0603	C27, C28, C29	CAP-0201	GRM1555C1H220JA01D	3
1uF	CAP CER 1UF 25V X5R 0603	C38, C39	Cap-0603	CC0603KRX5R6BB105	2
DNP	CAP CER 1PF 50V C0G/NP0 0402	C44, C45	Cap-0402	CC0402BRNPO9BN1R0	2
100pF	CAP CER 100PF 50V C0G/NP0 0402	C47	Cap-0402	CC0402JRNPO9BN101	1
0.01UF	SCHOTTKY DIODE 1MB 100V 5A	D1	RMB_	0.01UF	1
SMBJ33SB-M6	Independent 3W 33V SMB Zener Diodes ROHS	D2	RMB_	SMBJ33SB-M6	1
10MA/S830BT3G	DIODE ZENER 10V 0.5W SMA	D3	SMA	10MA/S830BT3G	1
TPSD04D03S5V	5450 ESD Protection Devices ROHS	D4	TPSD04D03S5V	TPSD04D03S5V	1
LESD5D5.0CT1G	SOD-523 ESD Protection Devices ROHS	D5	SOD-523	LESD5D5.0CT1G	1
LESD4BD3.3CAT5G	5V ±3V 3.3V Bi-Directional SOD-882 ESD Protection Devices ROHS	D6, D14	SOD-882 Bi Directional	LESD4BD3.3CAT5G	2
PESD5V0H1BSF/L	TVS DIODE 5V 9VAM 6.8VC DSN0603-2	D8, D13	DSN0603-2	PESD5V0H1BSF/L	2
LESD5D5.0CT1G	SOD-523 ESD Protection Devices ROHS	D9, D11	SOD-523 (Bi-Directional)	LESD5D5.0CT1G	2
ESD53D3F-3/TR	15A 20V 7V SI Bidirectional SOT-23 TVS ROHS	D10	Lce-23	ESD53D3F-3/TR	1
SM712	SOT-23 TVS ROHS	D12	SOT-23	SM712	1
SMD0603-050	5V 500mA 40A 3A 0603 Resettable Fuses ROHS	F1	Fuse-0603	SMD0603-050	1
100MHz	0190070.MXAD 220 OHM 0805 SMD	FB1	Ind-0805	MPZ2012S221A TD25	1
HC-1.25-12PLT	1x12P 12P ProtoBuddy(MX 1.25) Tin 12- 20°C~+80°C 2A 1.1.25mm Brass Surface Mount,Vertical SMD,P=1.25mm Wire To Board Connector ROHS	J1, J2	HC-1.25-12PLT	HC-1.25-12PLT	2
HC-1.25-6PLT	6x6P 6P ProtoBuddy(MX 1.25) Tin 6- 20°C~+80°C 2A 1.1.25mm Brass Surface Mount,Vertical SMD,P=1.25mm Wire To Board Connector ROHS	J3	HC-1.25-6PLT	HC-1.25-6PLT	1
U.FL-R-SMT-1(80)	CONN U.FL R/PT STK 50 OHM SMD	J4, J5	U.FL-R-SMT-1(80)_	U.FL-R-SMT-1(80)	2
WRAND-113	Copper Alloy Self Inomb Gold plating 1.35mm Deck - 40°C~+85°C 260°C NanoSIM SMD 5/M Card Connectors ROHS	J6	WRAND-113	WRAND-113	1
PDGD Connector 4 Pin	Plugin Pwr 2.54mm 4 Pin	J7	Pdgd-4Pin	PDGD Connector 4 Pin	1
TB008A-508-04BE	TERMINAL BLOCK, SCREW TYPE, 5.08	J8	TB008A-508-04BE	TB008A-508-04BE	1
691513WY-02H-C00D30	Gold 300V 3A Straight Square Pins 2.5mm 2P -200°C 6mm ~ 40°C ~+105°C 2mm 2.54mm Plugin Black Brass 1x2P Plugin,P=2.54mm Pin Headers ROHS	J9	HDR1X2_	691513WY-02H-C00D30	1
TB008A-508-03BE	TERMINAL BLOCK, SCREW TYPE, 5.08	J10, J11	TB008A-508-03BE	TB008A-508-03BE	2
PZ254V-11-07P	Straight Square Pins 2.5mm 7P 6mm ~ 40°C ~+105°C 2mm 2.54mm Plugin Black Brass 1x7P Plugin,P=2.54mm Pin Headers ROHS	J12	HDR1X7	PZ254V-11-07P	1
PZ254V-11-05P	Straight Square Pins 2.5mm 5P 6mm ~ 40°C ~+105°C 2mm 5 2.54mm 金色 Black Brass 1x5P Plugin,P=2.54mm Pin Headers ROHS	J13	HDR1X5_	PZ254V-11-05P	1
47uH	2.3A 47uH ±20% 1A 125mΩ SMD,10x11mm Inductors (SMD) ROHS	L1	MDA1040-470M	MDA1040-470M	1
68nH	2nd Chip Wirewound 68nH 5% 300MHz 17Ω Factor Ceramic 0504 SMD	L2	IND-0603(1608 Metric)	0603CS-68NXJLW	1
Yellow	Yellow-Green 572nm LED Indication - Discrete 2.05V 0603 (1608 Metric)	LED1	LED-0603	19-21/G6C-FM2N2B/3T	1
RED	Red 632nm LED Indication - Discrete 1.95V 0603 (1608 Metric)	LED2	LED-0603	19-21/R7C-AK1L2BY/3T	1
Red	LED-RED CLEAR CHIP SMD	LED3, LED4, LED5, LED6	LED-0402	APHDM-1BW-2	4
TEST POINT	Test Point DN	P1, P2, P3	Test-point	TEST POINT	3
MMBT2222A-7-F	TRANS NPN 40V 0.6A SMD SOT23-3	Q1, Q2, Q3, Q4, Q21	SOT-23	MMBT2222A-7-F	5
BCM7C,315	TRANS NPN 45V 0.1A TO-236AB	Q5, Q6, Q7, Q8, Q9, Q10, Q11, Q14, Q17, Q18	SOT-23	BCM7C,315	10
RLMU0402TRPBF	MOSFET P-CH 20V 4.7A SOT23	Q13, Q24	SOT-23	RLMU0402TRPBF	2
BCI857A,215	TRANS PNP 45V 0.1A TO-236AB	Q15, Q16	SOT-23	BCI857A,215	2
MMBT5551	TRANS NPN 160V 0.6A SOT-23	Q19, Q20	SOT-23	MMBT5551	2
0R	RES 0 OHM JUMPER 1/10W 0603	R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R96, R99	Res-0603	RC0603JR-070RL	21
10K	RES 10K OHM 5% 1/10W 0603	R20, R41, R43, R47	Res-0402	RC0402FR-0710KL	4
1K	RES 1K OHM 1% 1/10W 0603	R21, R22	Res-0201	RC0201FR-071KP	2
10R	RES SMD 10 OHM 1% 0.5W 0402	R23	Res-0402	CRCW040210R0FKED	1
220R	RES SMD 220 OHM 5% 1/10W 0603	R24, R27	Res-0603	CRCW0603220RFKEA	2
4.7K	RES 4.7K OHM 1% 1/10W 0603	R25, R26, R30, R37, R39, R40, R42, R90	Res-0402	RC0402FR-074K7L	8
47K	RES 47K OHM 1% 1/10W 0603	R28, R29, R31, R32, R38, R91	Res-0402	RC0402FR-0747KL	6
22R	RES 22 OHM 1% 1/10W 0603	R33, R34, R35, R36	Res-0201	RC0201FR-0722RL	4
0R	RES 0 OHM JUMPER 1/10W 0603	R44, R46, R93, R94	Res-0402	RC0402FR-070RL	4
120R	RES 120 OHM 1% 1/10W 0603	R45	Res-0603	RC0603FR-07120RL	1
1K	RES SMD 1K OHM 1% 1/10W 0603	R48, R49, R54, R55, R71, R73, R74, R75, R84, R85	Res-0603	RC0603FR-071K	10
4.7K	RES SMD 4.7K OHM 1% 1/10W 0603	R50, R51, R52, R53, R58, R59, R66, R69, R70, R72, R80, R81	Res-0603	RC0603FR-074K7L	12
330R	RES SMD 330R 1/10W 0603	R56, R57	Res-0603	RC0603JR-07330RL	2
100K	RES 100K OHM 1% 1/10W 0603	R60, R61, R62, R63	Res-0603	RC0603FR-07100KL	4
10K	RES SMD 10K Ohm 1% 1/10W 0603	R64, R65, R82, R83	Res-0603	RC0603FR-0710KL	4
47K	RES SMD 47K OHM 1% 1/8W 0805	R66, R67	RES-0805	CRCW080547K0FKEA	2
150K	RES 150K OHM 1% 1/10W 0603	R76, R78	Res-0603	CRCW0603150KFKEA	2
DNP	RES SMD 2M OHM 1% 1/10W 0603, RES 0 OHM JUMPER 1/10W 0603	R77, R79, R87, R89	Res-0603	RC0603FR-072M4L, RC0603JR-070RL	4
51R	RES 51 OHM 5% 1/4W 1206	R82	Res-1206	RC1206FR-0751RL	1
470K	RES 470K OHM 1% 1/10W 0603	R86	Res-0402	RC0402FR-07470KL	1
AP1509-33SG-13	IC REG BUCK 3.3V 2A SO-8	U1	SOP-8L	AP1509-33SG-13	1
LDS6100DPU33R	IC REG LINEAR 3.3V 1A SOT89	U2	SOT-89-3-(L-B-L-B)	LDS6100DPU33R	1
A7873E (FAB)	IC TIER MICRO 6.5-10.5A SMD	U3	A7873E (FAB)	A7873E (FAB)	1
7180S03DCUT	IC REG BUCK ADJ 3A 6TSOP	U4, U6	7180S03DCUT	7180S03DCUT	2
SMF05CT1G-65	TVS 5V Unidirectional 6V SOT-363/SOD Protection Devices ROHS	U5	SOT-363	SMF05CT1G-65	1
SP3485EIM/TR	SOP-8 RS-485/RS-422 5V ROHS	U7	SOP-8	SP3485EIM/TR	1